Activities to support dressing skills

- backward chaining: teach the last action first and work backwards. For example,
  stage one: put the sock on the child's foot
  child pulls up the sock;
  stage two: put the sock on the foot as far as the instep
  child pulls sock over the foot, etc.
  'You do the last little bit.'
- easy fastenings;
- clothes not too tight;
- clothes placed on the bed, as if on the body;
- clothes placed left to right in the order to put them on;
- clothes placed inside a hoop when undressing for PE;
- no fiddly fastenings – belts, laces, bows.

The occupational therapist may provide additional advice on specific difficulties that individual children may have with dressing.

Section 4: Handwriting and drawing

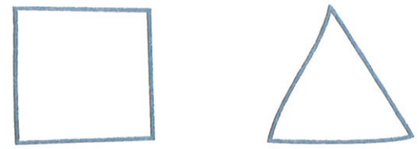
Handwriting is a fundamental motor skill for children to acquire and it is an example of maturationally controlled behaviour. This means that the motor skills which underpin writing follow a regular sequence of development which emerges as the child matures neurologically. Some learning is, of course, required but the learning cannot be significantly speeded up by coaching. The motor skills required for handwriting are, therefore, more linked to the neurological development of a child than to factors such as motivation, attitude or concentration. A negative attitude towards writing tasks may reflect underlying problems with the mechanics of handwriting which make the activity difficult and unrewarding for the child. The more difficult the mechanics of handwriting are for the individual, the more practice will be needed in order to master the skill but there is no direct relationship between how well most children write and how much practice they have had with writing.

The orderly progression of skill development is illustrated by the designs below:

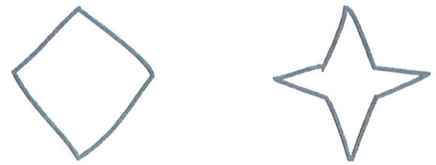
Group one

Group two

Group three



Group four



Children cannot copy or draw a later group until they have learnt the earlier groups. So any child who can draw group four will be able to draw all the designs in one, two and three. A child who cannot draw the designs in group one cannot draw any of the designs in two, three and four.

Groups one and two need to be mastered before letters are attempted and the order of difficulty of letters can be determined by consideration of the sequence. There should be more awareness of this in the teaching of handwriting.

If a five year old is not yet copying crosses and angles we know that they are going to find it more difficult than most children to master the writing of letters and numbers. Any or all of the following conclusions can be drawn:

- The child may well take longer to learn how to write them. They will compare themselves unfavourably with their peers.
  Action point: how can we protect their self-esteem and encourage them to keep trying and not give up?
- The child may need to be given the learning task in small doses. There are fewer numbers to learn than letters and once mastered the child can do quite a lot of independent number work.
  Action point: let's teach them numbers first.
- Some letters are more easily made than others.
  Action point: let's teach them these first, i.e. those based on a circle – o, c, a.
- Letters can be divided into families depending on how they are made. Above are the ones based on circles; others, for example, are based on a 'wiggly' pattern – m, n, w.
  Action point: let's teach the writing of the letters in these families.
- The most important early goal for a child is learning how to write their name.
  Action point: regardless of how difficult the letters are to make, let's give them lots of practice in the letters they need to write their name.
- Analyse the movement patterns the child has to make in order to write the letters.

Action point: can we help them by getting them to do the movements with their whole arm and using other materials such as sand, paint, etc.?

The list of possible developmental goal related strategies presents alternatives, some of which could be run in parallel but with others a choice needs to be made. Should I, for example, start with the child's name or with the letters based on O? The choice needs to be made at the time. It depends on how difficult the child finds it to draw crosses, what the letters are in the child's name and how many there are, how important writing their name is to the child, how willing the child is to struggle with something that they find difficult and many other questions. This approach recognises that although our development has much in common we are also unique individuals operating within unique combinations of circumstances.

Handwriting is a fundamental educational goal basic to progress through the school system. All ages of children spend over a third of their time at school handwriting (Alexander 1992). Five and six year olds spend thousands of hours practising to gain mastery over these complex motor patterns. A few individuals continue into the late primary and secondary phases of education with very poor handwriting skills. The curriculum demands involve an increase in the speed of writing which is required particularly at the point of secondary transfer and most children will write faster and faster until they leave formal education. Pupils' limitations in writing are underidentified in schools and teachers do not always realise how fast children need to write in lessons. They fail to get full notes, don't get down their homework, fail to cover all the points when making a written response in class and struggle to write enough in essays.

In this section we are considering the development of handwriting as a skill. Three broad phases can be identified: the early phase, the middle phase and the final phase.

Early phase

- Establishing a dominant hand and a support hand.
- Mastering the thumb and two finger grip.
- Learning the basic movement patterns for the letters and numbers.

Establishing a dominant hand and a support hand

By five years, children should have hand dominance. The orientation of symbols (letters and numbers) on a page is difficult to do without a dominant side; for example, the horizontal line in a number seven starts on the left and travels to the right, while in a number five it proceeds from the right to the left. Children with no difficulties but without clear dominance can reverse numbers, figures, words and whole sentences. Hand dominance is necessary to enable one hand to practise in a specialist way the complex patterns. There is also a role

for the support hand to learn in moving and holding the paper in position. Children at five years should be encouraged to use just one hand for writing. Left-handed children should experience no special difficulties providing they are given elbow room on tables and desks and provided that they adopt the correct paper position. The 'hook' left-hand grip is a simple result of the right-handed paper position being adopted. When writing, all the letters of the word being written need to be seen. Right handers can see what they are writing because the pen moves away from them. If a left hander uses the right-handed paper position the pen moves towards and across them. A left-handed writer should have the paper to the left and angled at 45 degrees to the vertical.

The best way to correct a left-handed grip that covers what is being written is to move the paper round gradually while holding the pencil further up the stem. Without guidance, by ten or eleven left handers have developed their own strategies, often the hook.

Mastering the thumb and two finger grip

Children begin using drawing and writing tools in their palms. It is usually adults who introduce them to the tripod grip. Initially, despite using a three finger grip, children will make the letter patterns by moving their hand and arm more than their fingers. Skilled writing involves finger and wrist movements with only a sliding action of the arm. In the early phase the grip is the important goal. A minority of children find the grip too difficult as it allows the pencil to wander freely unless good motor control is used. These children often adopt a grip that consists of placing the thumb and both fingers on the stem of the pencil. This variation is not as flexible but it serves the purpose. Poor hand coordination can lead to other unusual grips, for example, with the pencil set between the first and second fingers. After a period of helping the child adopt the correct position the

Box 4.6 *Writing left handed*

Rules round-up: ten-point plan

1. Select a writing surface and chair, suitable for your own height.
2. Sit towards the right of the desk or table, leaving plenty of space for writing on the left side of your midline.
3. Place the writing paper to the left of your body midline.
4. Tilt the paper up to 32 degrees in a clockwise direction.
5. Select a writing tool which moves smoothly across the paper.
6. Do not select a pen with a metallic and/or slippery barrel.
7. Support the paper with the right hand.
8. Position the writing tool in a 'below the line' position.
9. Keep the writing forearm parallel with the paper edge as you write.
10. Hold the writing tool sufficiently far from the point to ensure that you can see what you are writing.

Box 4.7 *Pencil control and pressure on the paper*
Some children experience tension in their fingers, wrists and/or forearms when they write and relaxation exercises may help with this.

If pressure on the paper is a major problem, the carbon paper 'sandwich' may be useful:

——————————— paper (1)
- - - - - - - - - - - - - - - carbon
——————————— paper (2)
- - - - - - - - - - - - - - - carbon
——————————— paper (3)

The aim of the task is for the child to write on paper (1) without marking paper (3). When phase one is achieved, remove paper (3) and the second carbon and concentrate on not marking paper (2).

child's preference should be followed. Indeed one of the leading experts on handwriting, Rosemary Sassoon (Sassoon and Briem 1993), argues that the pen held between the first and second fingers may well be a more appropriate position for modern pens. She counsels against rigid adherence to the tripod grip. This alternative grip was used more widely in the past, particularly in classical times. It is a firmer grip but less flexible.

Developing the grip is facilitated by teachers of young children in a number of ways:

- thicker than average pencils;
- triangular shaped thick pencils;
- soft plastic triangular pencil grips that slip on to standard pencil;
- moulded hard plastic grips that have indented thumb and finger positions, for standard pencils.

A triangular rather than a rounded stem encourages the correct grip. Children with poor control who have a tight grip with heavy pressure on the pencil usually benefit from the soft triangular grips while those with difficulties in learning the finger skills required benefit most from the moulded grips. Generally children should be allowed to experiment especially as six and seven year olds while assistance in helping the grip is continuing. Beyond seven years options are severely reduced. Sometimes elastic bands placed round a pen to thicken it can prove beneficial. A lump of Blu-tack is often more acceptable to older pupils and easy grip and self-grip pens have recently become available in the form of both biros and roller balls.

Learning basic movement patterns for letters and numbers

Learning the movement patterns is at the heart of the task. They are learnt initially in the whole arm and transferred down to the hand where in skilled writing they are exclusively located. The course

of this process can be observed in the changes in size of children's writing. Initial large writing is due to arm movements predominating while standard writing is produced by wrist and finger movements only. For most children their writing will gradually become smaller as control over the movement is transferred in stages to the fingers. This process is normally completed by seven or eight years. Sometimes children make a sudden switch from large writing to small writing. This is always a sudden switch from arm to hand control. It is usually found in poorly coordinated children who have found it hard to learn the movement patterns and occurs around seven. The best way to help children become skilled handwriters is to work on the movement patterns through whole arm movements while at the same time building the strength and flexibility of the hand and fingers through appropriate exercises and games. Clearly a third element remains straightforward practice at handwriting. We have listed strategies for helping the child's arm learn the movement patterns and strategies for hand and finger skills at the end of this section.

Teaching handwriting

- *The traditional British way:* At the infant phase, handwriting is the copying of individual letters, printing. At about seven a cursive script is taught whereby movement patterns that link the letters are added and complete movement patterns learned for words. Adult writers tend to retain some letter separations so that a full cursive hand is rare.
- *The alternative way:* In parts of the world such as continental Europe where children start school later, they learn a cursive script from the start.

Young children in Europe tend to print, which suggests that it is a separate phase of development when it is easier to separate each movement pattern to form individual letters. Movement patterns for whole words and to support spelling are better learned through a cursive script. In addition, handwriting is a flowing movement which is more efficiently executed as a continuous pattern. Handwriting experts generally recommend the teaching of cursive handwriting from the beginning but for children with poor movement control and for young children this may be a difficult target for them to achieve.

Middle phase

The middle phase of handwriting development begins within the first phase and partly parallels it.

- Transferring movement patterns fully into the hand.
- Learning to maintain an even size and to orientate the letters/words on a line, either real or imaginary.
- Introducing a range of tools.

Transferring movement patterns

We have already discussed some aspects of the transfer of movement patterns into the hand. This can be encouraged by providing lined paper when children reach the age of six years. The introduction of lined paper has to be carried out carefully. At first children need to write on plain paper so that they can produce the letters according to the size dictated by their arm and hand movements. The process of transferring handwriting skill into the hand varies between children hence the size of their writing varies. Lines impose size restrictions on children. During Year 1 and Year 2 children should move to lined paper as they choose. Paper with differing spaces between the lines should be available.

Learning to maintain size and orientation

Learning to maintain an even size, slope orientation and spacing of letters and words is also helped by lined paper but the points listed above still apply. At first the child's whole effort goes into producing the correct movement patterns. Plain paper is required in order to give full scope for the child's individual strategy. Gradually the child aligns the letters more carefully and, through practice, begins to observe size, constancy and consistent spacing. Lined paper, introduced as this later process is beginning, can help evenness of handwriting to develop. The strategy for helping children with difficulties in the primary years consists of emphasising the lines. Often a middle line is added to set the top height for small letters. For children with severe difficulties the three lines are coloured and some children require squared paper. The lined and squared paper needs to be prepared on the basis of the child's average letter size when writing on plain paper.

Introducing a range of tools

The pencil is the best tool to use for early handwriting though the movement patterns should be practised in a wide range of media – paint, sand, air tracing, chalk, etc. A pencil bites into the paper enough to absorb small unskilled movements of the hand and fingers. Only clear movements shift the pencil, hence cleaner lines are made and the control needed to obtain the desired result is easier. Historically the major transfer point was from the pencil to the ink pen around seven or eight. Pens come in all shapes and sizes and the child with a less orthodox grip will need to experiment with a range of pens to find one s/he can control comfortably. The best modern strategy is to move from the pencil to the fibre tip but ink pens remain an option for older children should they wish to transfer. Biros are the hardest to control as the roller ball at the tip gives very little bite and is free to register all small muscle movements in any direction. Its lack of bite does make it, however, the fastest writing instrument and it is ideal for skilled handwriters, though surprisingly difficult to keep even when writing at speed.

Final phase

The final phase of handwriting development involves the continual speeding up of handwriting. Dutton (1989) has described how this moves from an average of 1.5 words per minute at age five to 18 words per minute at sixteen. This improvement is largely a practice effect and speed gains continue to be made at secondary school. Speed of writing is a reflection of a deep skill and 'being a slow writer' who misses or avoids recording opportunities is a matter for support and assistance rather than criticism and social pressure. There is little that can be done about skill development in this area as the child's handwriting patterns and habits are well set before speed becomes an educational issue.

Handwriting and whole body movement come together in the context of the body position which is required to write. A stable base is needed with a chair a third of the height of the child and a table a half of the child's height. The surface should slope gently towards the child; rare nowadays. The paper should slope at the same angle as the child's arm. Once a poor writing posture is established from whatever cause, it is difficult to alter. This alone can affect writing speed, distort letters and lead to fatigue or pain to the writer.

Drawing

From a motor perspective drawing involves the ability to make the basic geometric shapes and is thus complete at six years. This goes some way to explain why some children who experience significant problems with handwriting are described by parents and teachers as 'good at drawing'. This is a form of written expression which they are able to perform and from which they may derive pleasure. A cartoon format is often favoured or the lines may be 'feathered' rather than clearly defined. This latter adaptation to accommodate poor pencil control can be attractive in drawing but is inappropriate for handwriting.

A major developmental goal centres on drawing a person, beginning with the face and head and moving into the periphery, hands and fingers, towards the end of the process. Drawing involves many aspects other than motor control and is therefore not dealt with in detail here.

Handwriting and drawing

Developmental milestones

15 mths	picks up small objects between finger and thumb;
1 yr 6 mths	scribbles, holding the 'crayon' in the palm of the hand;
2 yrs	imitates in drawing horizontal, vertical and circular strokes and scribbles in circles and straight lines;
2 yrs 6 mths	imitates a circle; imitates a V;

3 yrs	copies a cross, a T and an H;
3 yrs 6 mths	draws a person with head, eyes, nose, mouth and legs;
4 yrs	copies a Y; drawing of a person now includes a trunk;
4 yrs 6 mths	drawing of a person now includes arms;
5 yrs	can now copy all the letters including a K and E; copies a square and then a triangle;
6 yrs	drawing of a person now has hands;
6 yrs 6 mths	copies a diamond;
7 yrs	size of handwriting approaches the general standard.

Activities to support the development of handwriting

Activities to establish movement patterns

- air tracing beginning with circular movements;
- copy whole arm movement patterns on a chalk board;
- use a brush to paint employing all varieties of whole arm movement. Initially the child is learning about the trace that their arm movements leave. Use an easel. Paint all the letters in large brush strokes before drawing them. Painting is underused in the early stages of writing;
- talk through all movements – up, down, round, across, stop, start, circle, left, right, forwards and backwards, clockwise, anti-clockwise. Working on a vertical surface makes these directions easier for young children;
- parallel movement patterns with music and rhymes;
- draw a line through various mazes and pathways. These can require the whole range of movements – circular, wavy, zig-zag, sudden changes of direction, etc. Start with wide and then decrease to narrow pathways;
- dot-to-dot drawing going through simple continuous patterns to individual shapes.

Activities to develop wrist, hand and finger strength and skills

- finger and hand painting in any medium;
- finger rhymes putting firm pressure on the fingers;
- sticking coloured shapes on each finger and finger puppets;
- sand tracing with the fingers;
- palm and finger activity with dough, plasticine and clay;
- pull string, cord, wool, ribbon, etc. (wet and dry) through the fingers;
- all craft activities;
- pick-up-sticks game;
- use different strengths of peg on the side of a can;
- peg board patterns – eventually build in a race element;
- identifying objects in a feely bag;
- touch each finger in turn with the thumb. Start very slowly then turn into a race.